

JACKSON MORRIS

CURRICULUM VITAE

Simon Laufer Mathematical Science Institute

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EMPLOYMENT

Postdoctoral Researcher

Vanderbilt University

2027-2029

Postdoc SLMath

SLMath Semester Program

Motivic Homotopy Theory: Connections and Applications

Fall 2026

EDUCATION

Ph.D. in Mathematics

University of Washington

Advisors: Kyle Ormsby and John Palmieri

Thesis title: Cooperations in motivic homotopy theory

2021-2026

M.S. in Mathematics

University of Washington

2021-2023

B.S. in Mathematics, *magna cum laude*

University of Kentucky

2017-2020

RESEARCH INTERESTS

Motivic homotopy theory, chromatic homotopy theory, hermitian K-theory, algebraic K-theory, synthetic spectra, nilpotence and periodicity, Adams spectral sequence, periodic self-maps, redshift and blueshift

PUBLICATIONS AND PREPRINTS

10. *C₂-equivariant C-motivic stable stems*, joint with J.D. Quigley. In preparation.
9. *Diachromatic blueshift*, joint with Kyle Ormsby. In preparation.
8. *Motivic K-theory cooperations over the rationals*, joint with Sarah Petersen and Liz Tatum. In preparation.
7. *An exotic self-map of periodicity 1*. In preparation.
6. *Periodic phenomena in stable motivic homotopy theory*. Submitted. Available at <https://arxiv.org/abs/2607.18165>
5. *Immersion of C₂-projective spaces via Kℝ-theory*, joint with Manyi Guo, Alex Waugh, and Albert Jinghui Yang. Submitted. Available at <https://arxiv.org/abs/2604.25260>.
4. *Spectrum level splittings of truncated motivic Brown-Peterson cooperations algebras*, joint with Sarah Petersen and Elizabeth Tatum. Submitted. Available at <https://arxiv.org/abs/2509.19542>.
3. *Rings of cooperations for hermitian K-theory over finite fields*. To appear in **Algebraic & Geometric Topology**. Available at <https://arxiv.org/abs/2509.02786>.
2. *On the ring of cooperations for real hermitian K-theory*. To appear in **Annals of K-theory**. Available at <https://arxiv.org/abs/2506.16672>.
1. *Toric Double Determinantal Varieties*, joint with Alexander Blose, Patricia Klein, and Owen McGrath. **Communications in Algebra** 49 (2020), 7, 3085-3093.

SELECTED PRESENTATIONS & SEMINAR TALKS

Conference and Colloquium Talks

“Higher Witt K-theories”

AMS Sectional: Computational Homotopy Theory (Tempe, AZ)

November 2026

Young Topologists Meeting 2026 (Copenhagen, DK)

July 2026

Cascade Topology Seminar (Portland, OR)

May 2026

“Splittings and cooperations in motivic homotopy theory”

Joint Math Meetings 2026 (Washington, D.C.)

January 2026

“On the ring of cooperations for real hermitian K-theory”

European Talbot 2025 (Kolding, DK)

July 2025

Graduate Student Geometry & Topology Conference 2025 (Bloomington, IN)

April 2025

Joint Math Meetings 2025 (Seattle, WA)

January 2025

European Autumn School in Topology (Utrecht, NL)

September 2024

Motivic Homotopy, K-theory, and Modular Representations (Los Angeles, CA)

August 2024

“Finding Fixed Points with Invariants”

Reed College Mathematics Colloquium

February 2025

Seminar Talks

eCHT Research Seminar

September 2026

University of Washington Topology Seminar

February 2026

University of Illinois Urbana-Champaign Topology Seminar

January 2026

University of Virginia Topology Seminar

November 2025

University of Kentucky Topology Seminar

November 2025

Duke University Geometry and Topology Seminar

October 2025

University of Colorado Boulder Homotopy Theory Seminar

September 2025

University of Michigan Geometry Seminar

September 2025

University of Notre Dame Topology Seminar

September 2025

University of Washington Topology Seminar

June 2025

Contributed Talks

eCHT Reading Seminars

“An upper bound on the degree of symmetry of an exotic sphere”

March 2026

“Higher Adams differentials and hidden extensions”

November 2025

“Motivic twisted K-theory”

April 2024

Graduate Student Seminars

“Higher real K-theories, redshift, and blueshift”

May 2026

“Filtered objects, spectral sequences, and t -structures”

January 2026

“(A synthetic approach to detecting) v_1 -periodic families”

November 2025

“Algebraic and Milnor K-theory”

October 2025

“Galois descent and the Picard group of K-theory”

May 2025

“The Adams spectral sequence and Hopf algebroids”

February 2025

“Symplectic orientations”

August 2024

Workshops

“ $\pi_*\mathrm{THH}(\mathbb{Z}_p^{\mathrm{fil}})/p$ and $\pi_*\mathrm{THH}(\ell_p^{\mathrm{fil}})/(p, v_1)$ ” (Talbot, Nacogdoches TX)

August 2024

“Hochschild Homology” (European Autumn School in Topology, Utrecht, NL)

September 2024

SERVICE AND PROFESSIONAL ACTIVITIES

Refereeing and Reviewing

Algebraic and Geometric Topology

2025-Present

ZBMath

2026-Present

Seminar Organization

DUBTOP Seminar

2025-2026

Topics included: Chromatic homotopy theory, stacks, Goodwillie calculus

Back-To-School seminar Student lectures on fun math learned over the summer.	<i>Fall 2024</i>
Reading seminars Topics included: Algebraic K-theory, cohomology operations, motivic homotopy theory, vector bundles and topological K-theory, categorical homotopy theory, model categories	<i>2023-2024</i>

TEACHING EXPERIENCE

Instructor of Record

University of Washington Math 480 – Homotopy theory Math 208 – Linear algebra	<i>Spring 2026</i> <i>Winter 2023, Summer 2023</i>
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Graduate Assistant

Electronic Computational Homotopy Theory (eCHT) Research Seminar Quadratic Curve Counting	<i>2025-2026</i> <i>Fall 2024</i>
Park City Math Institute (PCMI) Experimental Math Lab	<i>Summer 2024</i>
University of Washington Courses included: Precalculus, Differential calculus, Multivariable calculus, Linear algebra, Introduction to proofs, Topology	<i>Fall 2021 - Spring 2026</i>

Directed Reading Mentor

University of Washington Topics included: Homotopy theory, Representation theory, Geometric group theory, Homological algebra	<i>2022-2025</i>
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Math Olympiad Judge

UW Math Hour Judge for elementary school math competition	<i>2023 - 2024</i>
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HONORS AND AWARDS

Lee and Cozette McFarlan Travel Fellowship University of Washington For travel to European Talbot in Kolding, DK.	<i>Summer 2025</i> <i>\$750</i>
Research Assistantship University of Washington For good research progress.	<i>Summer 2025</i>
JC Eaves Scholarship University of Kentucky For continued excellence in mathematics as a Kentuckian.	<i>2020</i> <i>\$2000</i>
Carolyn Bunyan Award University of Kentucky For mathematics majors interested in continuing education in mathematics.	<i>2020</i> <i>\$2000</i>

SELECTED WORKSHOP AND CONFERENCE PARTICIPATION

Extended Research Visits

SLMath Semester Program: Motivic Homotopy Theory (Berkeley, CA)	<i>Fall 2026</i>
PCMI: Motivic Homotopy Theory (Park City, UT)	<i>Summer 2024</i>

Conference Participation

Motivic Homotopy Theory: Applications and Connections (Berkeley, CA)	<i>October 2026</i>
AMS Sectional: Computational Homotopy Theory (Tempe, AZ)	<i>November 2026</i>
Algebraic Structures in Topology (San Juan, PR)	<i>July 2026</i>
Young Topologists Meeting (Copenhagen, DK)	<i>June/July 2026</i>
Cascade Topology Seminar (Portland, OR)	<i>May 2026</i>
(∞, ∞)-Categories Workshop (Chicago, IL)	<i>March 2026</i>
JMM 2026 (Washington, D.C.)	<i>January 2026</i>
Midwest Topology Seminar (Minneapolis, MN)	<i>October 2025</i>
European Talbot (Kolding, DK)	<i>July 2025</i>
WAGS Spring 2025 (Vancouver, CAN)	<i>April 2025</i>
GSTGC 2025 (Bloomington, IN)	<i>April 2025</i>
JMM 2025 (Seattle, WA)	<i>January 2025</i>
SLMath Hot Topics: Life after the Telescope Conjecture (Berkeley, CA)	<i>December 2024</i>
European Autumn School in Topology (Utrecht, NL)	<i>September 2024</i>
Talbot 2024 (Nacogdoches, TX)	<i>August 2024</i>
Motivic Homotopy, K-theory, and Modular Representations (Los Angeles, CA)	<i>August 2024</i>
eCHT Research Workshop on Hopf Rings (Online)	<i>June 2024</i>
JMM 2024 (San Francisco, CA)	<i>January 2024</i>
Scissors Congruence, Algebraic K-theory, & Trace Methods (Bloomington, IN)	<i>July 2023</i>
Cascade Topology Seminar (Vancouver, CAN)	<i>April 2023</i>
PRIMA Congress (Vancouver, CAN)	<i>December 2022</i>
Graduate Students in Geometry, Topology, and Algebra (Philadelphia, PA)	<i>May 2022</i>

REFERENCES

Mark Behrens John and Margaret McAndrews Professor of Mathematics University of Notre Dame <i>Relationship: Research mentor</i>	mbehren1@nd.edu
Kyle Ormsby Mathematics Department Chair Reed College <i>Relationship: PhD advisor and research collaborator</i>	ormsbyk@reed.edu
John Palmieri Professor of Mathematics University of Washington <i>Relationship: PhD advisor; Mathematics Department Chair during first 2 years of graduate study</i>	jpalmier@uw.edu
J.D. Quigley Assistant Professor of Mathematics University of Virginia <i>Relationship: Research mentor; Program supervisor and mentor at PCMI</i>	mbp6pj@virginia.edu
Charles Camacho Assistant Teaching Professor University of Washington <i>Relationship: Teaching supervisor and mentor</i>	camachoc@uw.edu
Natalie Naehrig Assistant Teaching Professor University of Washington <i>Relationship: Teaching supervisor and mentor</i>	naehrn@uw.edu